CHAPTER 1

INTRODUCTION

This chapter includes the general introduction of the system, problem statement, background information and objectives.

1.1 Background Information

The increasing demand for clean, sustainable, and uninterrupted energy has driven researchers and engineers to explore advanced renewable energy solutions. Among various green technologies, solar and wind energy play a significant role due to their abundance, low environmental impact, and long-term availability. However, the intermittent nature of these resources presents challenges when each source is used alone. Solar Photovoltaic (PV) systems depend entirely on sunlight, meaning their power output fluctuates with weather conditions, time of day, and seasonal variations. Similarly, wind turbines require consistent wind flow, and their power generation decreases significantly during low-wind periods.

To overcome these limitations, the concept of hybrid renewable energy systems has gained importance. A hybrid model combining a Vertical Axis Wind Turbine (VAWT) with Solar Photovoltaic (PV) panels provides a balanced and efficient solution. VAWT technology is particularly advantageous because it can operate at lower wind speeds, capture wind from any direction, and requires less space compared to horizontal-axis turbines. When integrated with solar panels, the combined system becomes capable of generating power throughout the day and night and across varying environmental conditions.

In the proposed hybrid system, Solar PV panels contribute maximum power during daylight hours, while the VAWT compensates for reduced solar output by harnessing wind energy, even during cloudy weather or at night. This complementary operation improves energy reliability, enhances overall system efficiency, and ensures continuous power generation. Such hybrid systems significantly reduce dependency on conventional fossil-fuel-based energy sources, thereby contributing to reduced carbon emissions and promoting environmental sustainability.

This model is particularly suitable for rural electrification, standalone home systems, small-scale industries, street lighting, and smart city infrastructure where stable and eco-friendly power is required. By integrating two renewable sources that naturally balance each other, the hybrid VAWT–Solar PV system represents an innovative and practical approach toward achieving energy security and supporting global clean energy goals.

1.2 Motivation

The main motivation behind developing a hybrid renewable energy system comes from the increasing need for clean, reliable, and affordable electricity. In many parts of the world, especially in rural and remote areas, people face frequent power cuts and do not have access to a stable electricity supply. At the same time, traditional energy sources like coal and diesel are causing serious environmental damage and are becoming more costly.

Solar energy and wind energy are natural resources that are freely available, but each source alone cannot provide constant power. Solar panels do not work at night or during cloudy weather, and wind turbines depend on the availability of wind. Because of these limitations, we felt the need to design a system that can continue producing electricity in all weather conditions.

This motivated us to create a hybrid model using a Vertical Axis Wind Turbine and Solar PV panel. By combining both sources, the system becomes more efficient, reliable, and suitable for real-world use. The motivation also comes from the desire to promote green technology and reduce pollution by using clean energy. Building this project gives us a practical understanding of renewable energy systems and encourages the adoption of sustainable practices for a better future**.**

1.3 Problem Statement

The increasing global demand for electricity, coupled with the rapid depletion of fossil fuels and rising environmental pollution, has created an urgent need for clean, reliable, and sustainable energy sources. Conventional solar and wind systems are often limited because solar power depends on sunlight availability, while wind power varies with wind speed and direction.

This causes instability in power generation, especially in rural or remote areas where continuous electricity supply is essential. To address these limitations, there is a need for

a **hybrid renewable energy solution** that can generate power in all weather conditions by combining complementary sources. A **Vertical Axis Wind Turbine (VAWT)** can produce energy even at low and variable wind speeds, while **Solar Photovoltaic (PV)** panels efficiently convert sunlight into electricity during the day. Integrating both systems can ensure steady, uninterrupted, and eco-friendly power output.

Additionally, for small-scale applications, there is a requirement for a **portable and compact hybrid system** that can be easily installed, transported, and used in locations without grid access—such as agricultural fields, outdoor events, rural homes, camping sites, and emergency situations

1.4 Report Outline

This report is organized into different chapters to clearly explain the work carried out in the project. It begins with an introduction about the need and aim of the system, followed by a brief review of related work. The system design, hardware descriptions are explained in detail in the further chapters. The results obtained from the solar and wind turbine and the discussion on system performance are also included. The report concludes with advantages, applications and the future scope of the project.

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1.5 Objectives

- To design a hybrid renewable energy system combining Solar Photovoltaic (PV) and Vertical Axis Wind Turbine (VAWT).
- To ensure continuous and reliable power generation under varying environmental conditions.
- To improve the overall efficiency of renewable energy systems by utilizing two complementary sources.
- To develop a compact, cost-effective, and low-maintenance power generation model suitable for small-scale and rural applications.

CHAPTER 2

LITERATURE SURVEY

2.1 General Introduction

A literature survey is carried out to study the Hybrid renewable energy systems have gained significant attention in recent years due to the rising need for clean, reliable, and sustainable power. Solar Photovoltaic (PV) and Wind Energy systems are widely used because they are environmentally friendly and abundant. However, each source has limitations when used alone—solar panels depend on sunlight, while wind turbines rely on wind availability. To overcome these challenges, researchers have proposed hybrid models that integrate both solar and wind energy sources. The combination improves reliability, enhances power generation efficiency, and ensures continuous energy supply under different environmental conditions.

2.2 Review of past work done

1: Solar–Wind Hybrid Systems

Previous research highlights that combining solar and wind systems increases energy availability. Many studies show that solar panels generate maximum power during the day, while wind energy often peaks during evening or cloudy conditions, making them complementary sources. – A Review Anil Tekale, Vaibhav Ware, Vishal Devkar, Ganesh Dungahu 2018

2: Vertical Axis Wind Turbines (VAWT)

Researchers found that VAWTs perform better in low-wind environments and are effective in urban or rural areas with unpredictable wind directions. Their ability to start at low speeds makes them suitable for hybrid systems. Swarupa Pinninti 2019

3: Hybrid Energy for Rural Electrification

Several papers demonstrate successful implementation of hybrid solar–wind systems in remote villages. These models ensure reliable power where grid supply is weak or unavailable. They also reduce dependency on diesel generators and lower operational costs.- Chetan Sonawane, Manav Velani, Akash Singh, Vikaskumar Tripathi 2017

4: Use of Charge Controllers and Battery Storage

Past work emphasizes the importance of charge controllers for regulating voltage from both sources. Battery storage systems are widely used to supply continuous power even when solar radiation or wind speed is low - Jaymin Pareshkumar Shah(published in IJSAT journal) 2023

5: System Efficiency and Cost Optimization

Earlier studies show that hybrid systems improve overall efficiency and reduce energy cost over time. Integration of inverter technology further enhances power quality, making the system suitable for household and small-scale applications.- Review Vaibhav Bhagat, Vipul Gedam, Tejas Ingle, Priti Kulmethe, Yash Jadhao, Manjusha Hedau2023

2.3 Summary

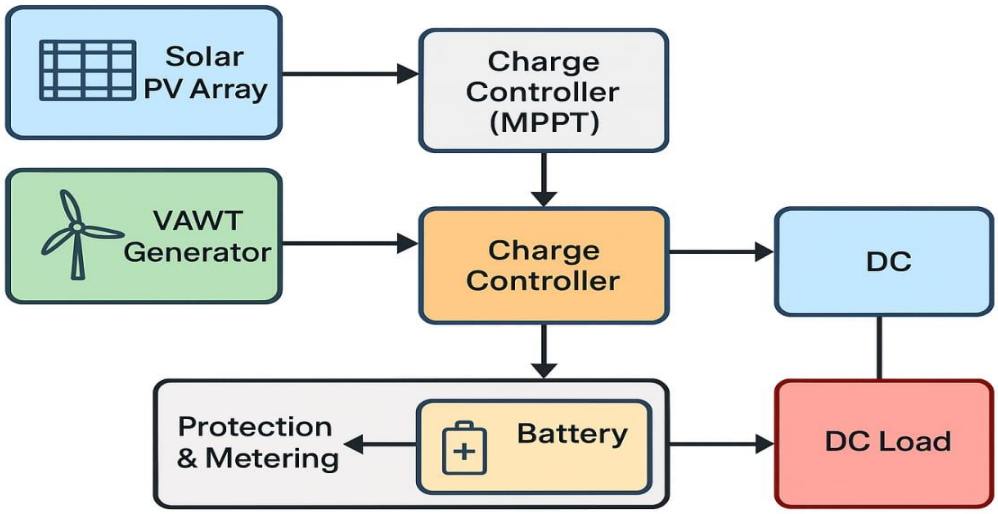
The literature shows that hybrid renewable energy systems combining Solar PV and VAWT technology offer a reliable and efficient solution for continuous power generation. Solar panels and VAWTs complement each other in different weather conditions, resulting in improved system stability and reduced energy fluctuations. Past research also supports the use of charge controllers, inverters, and battery storage for efficient power management. Overall, existing studies indicate that hybrid green power models are cost-effective, environmentally friendly, and ideal for small-scale and off-grid applications.

CHAPTER 3
SYSTEM DESCRIPTION

3.1 General Introduction

The hybrid renewable energy system combines a Solar Photovoltaic (PV) panel and a Vertical Axis Wind Turbine (VAWT) to generate continuous and reliable electricity. The solar panel converts sunlight into electrical energy during daytime, while the VAWT captures wind energy and produces power even at low or changing wind speeds. Both energy sources are connected to a charge controller, which regulates and manages the power flow to a battery for storage. From the battery, the stored energy is supplied to the load either directly (DC supply) or through an inverter (AC supply). By integrating two complementary renewable sources, the system ensures stable power generation in all weather conditions and improves overall energy efficiency.

3.2 Block Diagram and Description



1.Solar PV Array

Converts sunlight into DC electrical power.

Works efficiently during daytime and clear weather.

2. VAWT (Vertical Axis Wind Turbine) Generator

Converts wind energy into electrical energy.

Works in low and variable wind speeds; no need for turbine orientation.

3. Charge Controller (MPPT)

Regulates and balances power from both Solar PV and VAWT.

Ensures maximum power extraction using MPPT technology.

4. Battery (Energy Storage)

Stores the energy generated from solar and wind sources.

Provides continuous power supply to loads even when generation is low.

5. Protection & Metering

Includes fuses, circuit breakers, and surge protection.

Measures voltage, current, battery health, and overall system performance.

6. DC Output / DC Load

Final usable power output of the system.

Supplies DC loads such as LED lights, small DC motor, fans, sensors, et

3.3 Working Steps of the System

The solar panel and VAWT turbine generate electricity from sunlight and wind. This power is regulated by the charge controller and stored in the battery, which then provides continuous DC supply to the load.

The given diagram represents a hybrid renewable energy generation system that integrates a Solar Photovoltaic (PV) array with a Vertical Axis Wind Turbine (VAWT) generator to produce a stable and continuous DC power supply. The purpose of combining two different renewable sources is to ensure that electricity can be generated under varying environmental conditions—solar power during daytime and windy conditions, and wind power even at night or during low-sunlight hours. This makes the system more reliable, efficient, and suitable for off-grid or remote applications.

The system begins with the Solar PV array, which converts sunlight into direct current (DC) electricity. Since the power output of solar panels changes with sunlight intensity, angle, and temperature, the generated energy is not stable or optimized in its raw form. Therefore, the solar output is fed into an MPPT-based charge controller. MPPT (Maximum Power Point Tracking) technology continuously adjusts the operating point of the solar panel to ensure it always produces its maximum possible power. This controller regulates the voltage and current, prevents overcharging, and ensures safe power transfer to the battery and the load.

Parallel to the solar source, the system includes a Vertical Axis Wind Turbine (VAWT) generator, which produces electrical energy from wind. Unlike traditional horizontal turbines, a VAWT can operate efficiently in low or changing wind directions and is suitable for urban or low-space environments. The VAWT generator output, which may vary significantly due to changes in wind speed, is directed into its own charge controller. This controller conditions the wind-generated electricity by regulating voltage, converting AC to DC if needed, and ensuring that the generated power is safe for both the battery and the load.

Both the solar MPPT controller and the wind controller feed power into the central battery unit, which acts as the main energy storage component of the system. The battery stores excess energy when solar or wind power is abundant and supplies power when both sources are unavailable—such as during nighttime with no wind or during cloudy conditions with low wind speeds. The battery ensures continuous operation of the load without interruption. To protect the battery and other system components, there is a Protection & Metering block, which continuously monitors voltage levels, current flow, battery state-of-charge (SOC), temperature, and overall system health. It also provides protection against overcharging, deep discharge, short circuits, and reverse polarity, which helps to increase the battery life and maintain system safety.

The regulated power from the charge controllers and the battery is then supplied to the DC conversion stage, which ensures that the required and stable DC voltage is delivered to the load. This stage may include DC–DC converters to step up or step down the voltage depending on the load requirements. Finally, the processed and stabilized power is delivered to the DC Load, which may include lighting systems, small appliances, communication devices, or any DC-powered equipment. The load always receives a consistent and reliable power supply because it can draw energy directly from the renewable sources when available or from the battery when the sources are insufficient.

Overall, this hybrid system ensures maximum utilization of natural resources by combining two complementary renewable sources—solar and wind. The charge controllers maintain efficiency, the battery provides energy storage for backup, and the protection system ensures safe operation. The result is a reliable, efficient, and sustainable energy system suitable for off-grid applications, rural electrification, and small-scale renewable projects..

Key Features

- Dual renewable sources (Solar + Wind) for continuous operation.
- Works in all weather conditions.
- Reliable power supply for off-grid areas.
- Eco-friendly and low maintenance.
- VAWT runs at low wind speeds and any direction.
- Reduces dependency on fossil fuels.
- Supports DC and AC loads**.**

3.4 Summary

The hybrid VAWT–Solar PV system efficiently combines wind and solar energy to produce reliable, clean, and continuous power. It uses a charge controller and battery storage to ensure stable energy output in varying weather conditions. This system is ideal for homes, rural areas, farms, industries, and smart city applications. With improved efficiency, low maintenance, and environmental benefits, it provides a sustainable and effective renewable energy solution**.**

CHAPTER 4

HARDWARE SPECIFICATION

4.1 Introduction

The hardware components used in the hybrid renewable energy system play a crucial role in achieving efficient and continuous power generation. The model integrates both solar and wind energy sources, requiring reliable and compatible hardware to convert, store, and supply electrical power. Proper selection of components ensures stable operation, long-term durability, and enhanced performance of the hybrid system. The following section describes the hardware specifications and details of each component used in the model.

4.2 Hardware Requirements

Sr. No.	Component	Specifications	Purpose
1	Solar Photovoltaic (PV) Panel	10W / 20W / 50W, Poly/Mono-crystalline	Converts sunlight into DC electrical energy
2	Vertical Axis Wind Turbine (VAWT)	2–4 blades, Low wind speed operation	Captures wind energy from any direction and rotates the generator.
3	Charge Controller	12V, 10A/20A Hybrid controller	Regulates voltage, prevents battery overcharging, manages power from solar & wind
4	Battery Storage	12V, 7Ah/12Ah Lead-acid/Li-ion	Stores generated energy for continuous power supply
6	Stepper Motor	Voltage: 5V / 12V / 24V	Provides precise rotational motion in steps
7	Connecting Wires & Cables	Copper wires 1 sq.mm / 2.5 sq.mm	Transfers power between all components with minimum loss.

Table 1.1: Hardware Requirements of System

4.3 Description of Components

4.3.1 Solar Pannel

- Converts sunlight into DC electrical energy using photovoltaic (PV) cells.
- Typically made of polycrystalline or monocrystalline silicon cells.
- Designed to charge **12V batteries** in small hybrid and standalone systems.
- Provides stable power output during daytime under adequate sunlight.
- Lightweight, durable, and requires very low maintenance.
- Commonly used in home lighting systems, small renewable projects, and portable setups.
- Acts as the primary energy source in daytime for hybrid solar–wind systems.



4.3.2 Battery

- Regulates the voltage and current coming from solar panels and wind turbines.
- Prevents battery overcharging and deep discharging.
- Ensures safe and efficient charging of the battery.
- Protects the system from overvoltage, short circuits, and reverse current flow.
- Maintains battery health and increases battery lifespan.



4.3.3 Charge Controller

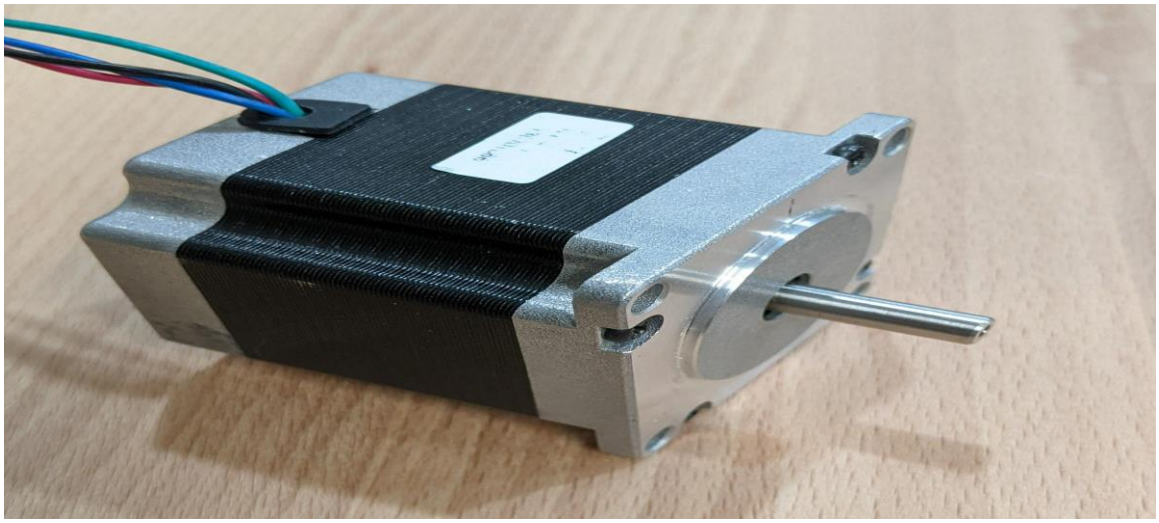
- Regulates the voltage and current coming from solar panels or wind turbines.
- Protects the battery from overcharging and deep discharging.
- Ensures stable and safe charging of the battery.
- Prevents reverse current flow from battery to the solar panel at night.
- Increases battery life by maintaining proper charging cycles.

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4.3.4 Stepper Motor

- Converts electrical pulses into precise mechanical movement.
- Rotates in fixed, equal steps (e.g., 1.8° per step).
- Allows accurate control of position, speed, and direction.
- Commonly used in robotics, CNC machines, 3D printers, and automation.
- Requires a motor driver to operate properly.



4.3.5 Connecting Wires

- Good quality wiring reduces noise & improves readings

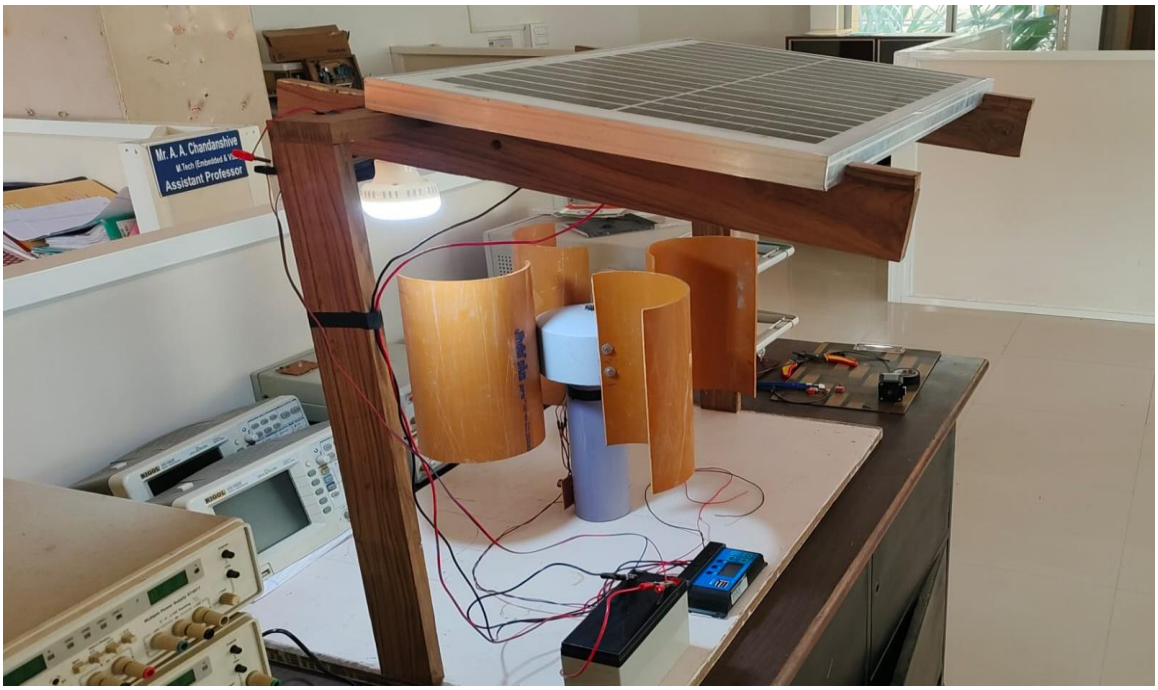


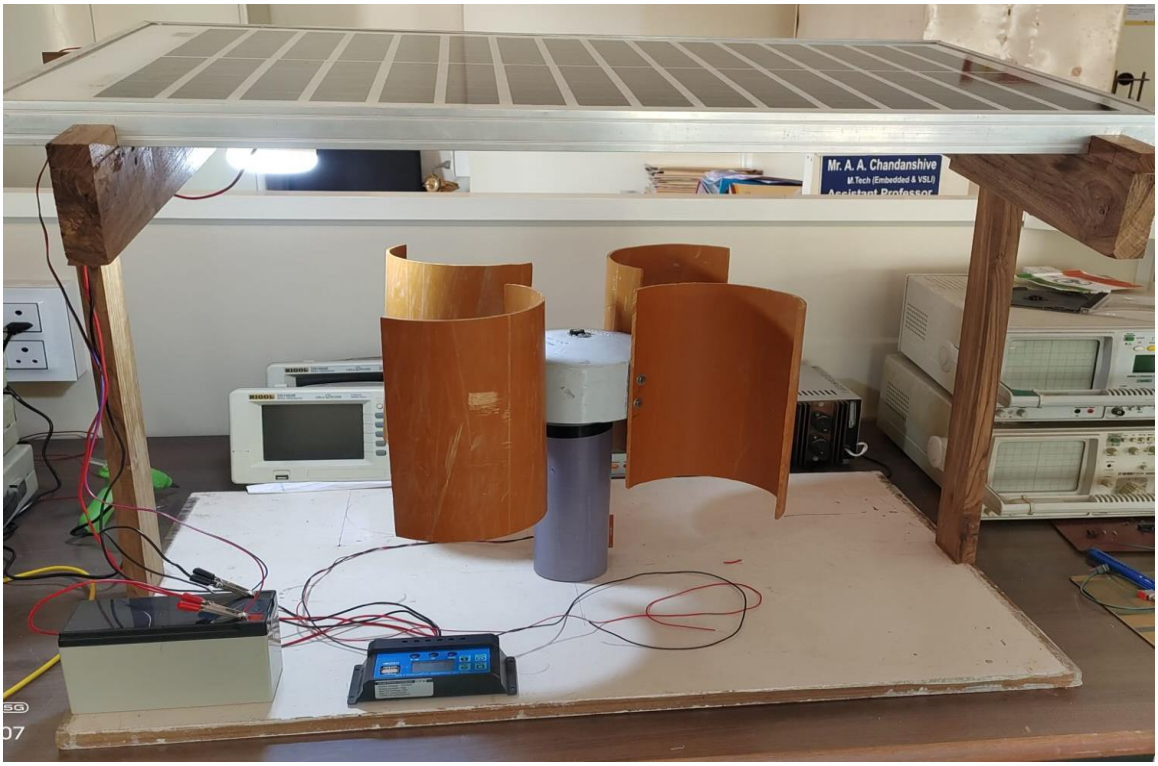
4.4 Summary

The hardware of the hybrid renewable energy system consists of key components such as the solar panel, vertical axis wind turbine, charge controller, battery, inverter, and monitoring circuits. Each component performs a specific function to ensure efficient and continuous power generation. The solar panel converts sunlight into electrical energy, while the wind turbine generates power from wind at varying speeds. The charge controller manages the power flow from both sources and protects the battery from overcharging or deep discharge. The battery stores the generated energy, and the inverter converts stored DC power into AC for operating household loads. Additional components like stepper motors, sensors, and wiring ensure smooth operation, improved control, and reliable performance of the hybrid system.

CHAPTER 5

RESULTS AND DISCUSSION




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M.Tech (Embedded & VSLI)

CHAPTER 6

BUSINESS PLAN AND COMMERCIAL ASPECT

The hybrid VAWT–Solar PV system offers a profitable and sustainable business opportunity by providing continuous clean energy. It has strong commercial potential in homes, farms, industries, and government projects, supported by growing demand and renewable energy policies

7.1 Market Opportunity

- Growing demand for clean and renewable energy due to rising pollution and energy shortages.
- Increasing electricity costs push customers to adopt alternative power solutions.
- Large potential in rural areas, remote villages, farms, and off-grid locations where grid power is unreliable.
- Government support through subsidies, renewable energy policies, and tax benefits.
- Rising interest from industries, institutions, and households in hybrid solar–wind systems for 24/7 power.
- Market value of hybrid renewable energy systems is expected to grow rapidly due to energy security needs and climate change commitments.

7.2 Target Customers

- Rural households needing continuous power.
- Farmers for water pumps, lighting, and agricultural equipment.
- Small industries and workshops seeking backup or primary power supply.
- Educational institutions and commercial buildings adopting green energy.
- Off-grid sites like telecom towers, remote stations, and construction sites.
- Government and NGOs working on rural electrification projects.
- Residential societies wanting hybrid rooftop installations.

- Smart city projects focusing on sustainable development.

7.3 Cost Estimation

Component	Approx. Cost (₹)
Solar Pannel	1800
Charge Controller	400
Battery	900
Stepper Motor	700
Wooden Structure	3000

Table 2.1: Cost Estimation

The system can be sold commercially at **₹15,000/- – ₹18,000/-** with profit margin.

7.4 Business Model

1.Product-Based Model

Selling complete hybrid energy systems (VAWT + Solar PV + Controller + Battery + Inverter) as ready-to-use packages.

2. Service-Based Model

Providing installation, maintenance, repair services, and Annual Maintenance Contracts (AMC).

3. Build–Own–Operate Model

Company installs the system at customer sites and charges monthly fees or per-unit energy usage.

4. Custom Solutions

Designing customized hybrid systems for farms, schools, community centers, and commercial buildings.

7.5 Future Commercial Scope

- Increasing global push for net-zero emissions boosts commercial demand.
- Scope for rooftop hybrid installations in cities.
- Use of IoT-based monitoring systems to improve energy optimization.
- Opportunity to integrate advanced batteries and storage solutions.
- Strong potential in portable hybrid energy systems for camping, military, and emergency services.
- Growing interest from companies adopting ESG and sustainability goals.
- High future demand from EV charging stations powered by hybrid renewable systems.

7.6 Summary

The hybrid VAWT–Solar PV system offers a powerful business opportunity in today's clean energy market. It provides reliable 24/7 electricity, reduces dependency on fossil fuels, and supports sustainable development. With a growing need for renewable energy in rural and urban areas, this system presents strong market potential. The business can target households, farms, industries, and government sectors while offering products, installations, and long-term service models. Future commercial opportunities include smart cities, IoT-based systems, and large-scale hybrid microgrids. Overall, the hybrid system is a scalable, profitable, and environmentally responsible business solution.

CHAPTER 7
CONCLUSION

The hybrid power generation model combining a **Vertical Axis Wind Turbine (VAWT)** and **Solar Photovoltaic (PV)** system provides a reliable and continuous source of renewable energy. By utilizing two complementary sources—solar energy during the day and wind energy at varying speeds—this system overcomes the limitations of single-source renewable setups. It ensures higher efficiency, improved power stability, and better utilization of natural resources. The hybrid model is suitable for both rural and urban applications, contributing to reduced pollution and enhanced sustainability.

8.1 Advantages

The main advantages of this system are:

- Provides continuous power generation in varying weather conditions.
- Enhances overall system efficiency compared to a single renewable source.
- Reduces dependence on fossil fuels and decreases carbon emissions.
- VAWT works even at low and multi-directional wind speeds.
- Solar PV offers a clean, low-maintenance electrical output.
- Suitable for remote areas where grid supply is limited.
- Reduces battery charging time due to dual input sources

8.2 Disadvantage

The disadvantages of this system are as given below :

- Initial installation cost is higher than single-source systems.
- Requires more space to install both PV panels and wind turbine.
- Energy output depends on environmental conditions such as sunlight and wind speed.
- Mechanical parts of VAWT may need regular maintenance.

- Power fluctuates if both wind and solar conditions are poor.

8.3 Conclusion

The hybrid system combining a Vertical Axis Wind Turbine and Solar Photovoltaic panel provides a reliable and efficient source of renewable energy. By using both wind and solar power, the system ensures continuous electricity generation in different weather conditions. It reduces dependency on fossil fuels, lowers pollution, and offers a sustainable solution for rural and urban power needs.

8.4 Future Scope

- Integration with IoT and smart monitoring systems to improve efficiency.
- Development of advanced lightweight materials for better turbine performance.
- Use of MPPT (Maximum Power Point Tracking) to optimize solar and wind energy extraction.
- Adoption in smart cities, rooftop systems, and portable hybrid power stations.
- Research on energy storage technologies such as supercapacitors and advanced Li-ion batteries.
- Possibility of grid integration to supply excess power to public networks.
- Improved turbine designs for ultra-low wind speed operation.

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ANNEXURE I

PROJECT CO-PO MAPPING

Course Outcome (COs)	
Student will be able to:	
C3.14.1	Demonstrate the ability to interface hardware such as soil sensors, RS-485 communication modules and Raspberry Pi for real-time soil monitoring.
C3.14.2	Develop and execute Python programs for acquiring, processing and displaying soil parameter data including NPK, pH, moisture and electrical conductivity.
C3.14.3	Analyze soil readings to evaluate soil health condition and support decision-making for suitable crop and fertilizer management.

Sr.No.	Program Outcome (POs)
P01	**Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems
P02	**Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
P03	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations
P04	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
P05	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
P06	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and

	the consequent responsibilities relevant to the professional engineering practice.
P07	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
P08	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
P09	**Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
P10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions
P 11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
P 12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

CO-PO Mapping Table

Course Outcome (COs)	**Program Outcome (POs)**											
	P01	**P02**	**P03**	**P04**	**P05**	**P06**	**P07**	**P08**	**P09**	**P10**	**P11**	**P12**
C3.14.1	**3**	**2**	**1**		**2**				**2**			**1**
C3.14.2	**3**	**2**	**3**		**2**	**2**	**2**	**2**	**2**		**1**	**3**
C3.14.3	**3**		**2**	**1**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**3**

The above Table shows the minimum mapping of CO with PO for Project